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Room-temperature magnetic order on zigzag edges of narrow graphene nanoribbons

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The possibility that non-magnetic materials such as carbon could exhibit a novel type of *s*–*p* electron magnetism has attracted much attention over the years, not least because such magnetic order is predicted to be stable at high temperatures[1]. It has been demonstrated that atomic-scale structural defects of graphene can host unpaired spins[2,3], but it remains unclear under what conditions long-range magnetic order can emerge from such defect-bound magnetic moments. Here we propose that, in contrast to random defect distributions, atomic-scale engineering of graphene edges with specific crystallographic orientation—comprising edge atoms from only one sub-lattice of the bipartite graphene lattice—can give rise to a robust magnetic order. We use a nanofabrication technique[4] based on scanning tunnelling microscopy to define graphene nanoribbons with nanometre precision and well-defined crystallographic edge orientations. Although so-called 'armchair' ribbons display quantum confinement gaps, ribbons with the 'zigzag' edge structure that are narrower than 7 nanometres exhibit an electronic bandgap of about 0.2–0.3 electronvolts, which can be identified as a signature of interaction-induced spin ordering along their edges. Moreover, upon increasing the ribbon width, a semiconductor-to-metal transition is revealed, indicating the switching of the magnetic coupling between opposite ribbon edges from the antiferromagnetic to the ferromagnetic configuration. We found that the magnetic order on graphene edges of controlled zigzag orientation can be stable even at room temperature, raising hopes of graphene-based spintronic devices operating under ambient conditions.

The intrinsic magnetism of graphite has a long and controversial history[1]. The origin of the measured magnetic signal is generally attributed to atomic-scale structural defects locally breaking the sub-lattice balance of the bipartite hexagonal lattice[5,6]. However, unambiguous identification of the structural sources of the measured magnetic signal has proved challenging because the sources are buried inside the bulk of the material. The isolation of single graphene layers[7] opens up new prospects in this direction[8,9] because their atomic structure is fully accessible for imaging and controlled modification. In particular, the graphene edges of a specific crystallographic orientation comprising carbon atoms from only one sub-lattice of the bipartite hexagonal lattice (zigzag) are predicted to host magnetic order[10], in striking contrast to graphene edges that incorporate an equal number of carbon atoms from both sub-lattices (armchair).

The strong influence of edge orientation on the electronic structure of graphene nanoribbons was predicted[11] long before graphene was isolated[7]. However, a systematic experimental verification of this fundamental prediction is still lacking, owing to very limited control over the edge orientation of the graphene ribbons investigated. Opening of a gap inversely proportional to the ribbon width has been experimentally demonstrated by electrical transport measurements[12,13]; however, no evidence on the effect of edge orientation has been revealed. This is attributed mainly to the random edge orientations and the presence of a considerable, nanometre-scale edge roughness, suppressing orientation effects[14]. Scanning tunnelling microscopy (STM) study of irregularly shaped graphene ribbons revealed that structures with more zigzag edges display smaller bandgaps than those with more armchair edges[15]. This is clearly indicative of edge-specific physics; however, the lack of orientation control did not allow a more systematic insight. In theory, there is a broad consensus that graphene nanoribbons with armchair edges are semiconductors, owing to the quantum confinement of the charge carriers, while zigzag ribbons host peculiar flat-band edge states[10,16]. In the picture neglecting electron–electron interactions, these edge states localized near the Fermi level render all zigzag graphene nanoribbons metallic. However, such one-dimensional metallic edge states with a high local density of states at the Fermi level become unstable upon electron–electron interactions. To lower the energy of the system, it is favourable to split the flat band (open a gap) by ordering the spins along the two ribbon edges with antiferromagnetic coupling between opposite edges. Consequently, the emergence of magnetic order is closely linked to altering the electronic structure of the ribbons, through opening a bandgap in the otherwise metallic zigzag nanoribbons[10,16,17]. This enables us to detect the signature of edge magnetism on individual graphene nanostructures by investigating their electronic structure (directly measuring magnetic signals would require a macroscopic amount of such ribbons). This edge-state magnetism and the associated bandgap opening in zigzag ribbons is consistently predicted by various theoretical models, including first principles density functional theory[16], and mean-field-theory-based Hubbard[18] and quantum Monte Carlo calculations[19], indicating that edge magnetism is a robust property of zigzag graphene nanoribbons, not sensitive to the specific details of the models. However, the stability of the magnetic order on real graphene edges and experimental conditions is strongly debated. Experimental indications that edge magnetism can indeed occur at low temperatures (7 K) have been provided by tunnelling spectroscopy measurements on graphene ribbons obtained by unzipping carbon nanotubes[20]. However, the random orientation of the edges and the influence of a possible strong edge–substrate hybridization[21] did not allow edge magnetism in graphene to be properly characterized. Although the magnetic order is expected to persist to some extent on zigzag segments of randomly oriented graphene edges, the mixing of different edge types is expected to substantially weaken the effect[19,22]. Therefore, the lack of experimental control over edge orientation seems to be one of the main reasons that the magnetic graphene edge states consistently predicted by various theoretical models have remained experimentally so elusive.

To realize graphene nanoribbons with precisely controlled crystallographic edge orientations, we used a nanofabrication technique based on STM[4]. In atomic-resolution images, the crystallographic directions of the graphene lattice can be identified and matched with the desired cutting directions. The direct cutting of graphene is done by beam-induced chemical etching activated by the sub-nanometre-wide channel of tunnelling electrons, locally breaking the carbon–carbon bonds

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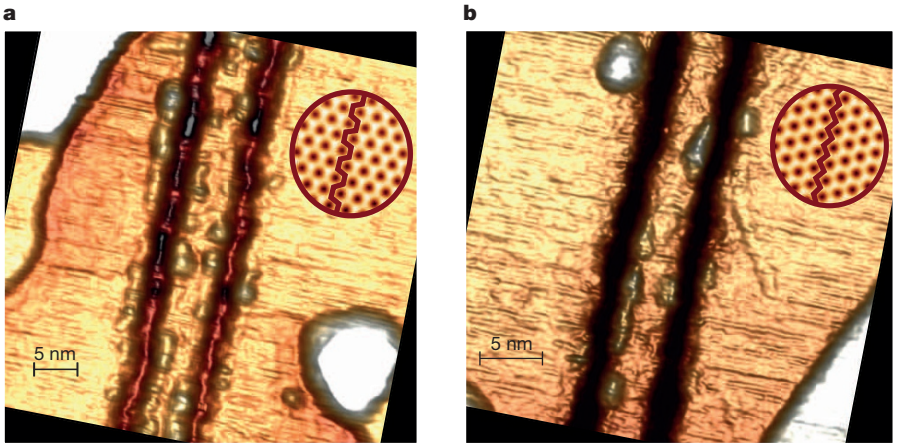

Figure 1 | Fabrication of graphene nanoribbons with precisely defined crystallographic edge orientations. STM image (500 mV, 0.8 nA) of a 5-nm-wide graphene nanoribbon with armchair edge orientation (**a**) and STM image of a 6.5-nm-wide ribbon with edges of precisely zigzag orientation (300 mV, 2 nA) (**b**) patterned by scanning tunnelling lithography in a graphene sheet deposited on a Au(111) substrate. The circular insets show atomic-resolution STM images, confirming the crystallographic directions of the edges. The atomic-resolution images in the insets were Fourier filtered for clarity. The protrusions on the otherwise highly regular edges are imaging instabilities.

underneath the atomically sharp STM tip apex. More details on the nanolithographic process are given in the Methods.

Using the STM nanolithographic technique, we defined nanoribbons into graphene sheets grown by chemical vapour deposition and transferred onto gold (Au) substrates with large Au (111) terraces. Au (111) has been predicted to be one of the few substrates preserving edge magnetism in supported graphene ribbons[21]. Nanoribbons with pre-defined armchair or zigzag edge orientation and widths ranging down to 3 nm were defined (Fig. 1). The as-fabricated ribbons display regular edges of subnanometre roughness. The bright protrusions near the edges in Fig. 1 are due to a known imaging instability[23], rather than edge defects or impurities. Upon optimizing the STM imaging, we found the edges defined by STM lithography to be straight and close to atomically smooth, free of detectable impurities, reconstructions or curvature (Extended Data Fig. 1).

Tunnelling spectra acquired on armchair ribbons displayed semiconducting current–voltage (I–V) characteristics, in contrast to spectra taken on graphene outside the ribbons (see Extended Data Fig. 2). The measured bandgap as a function of the corresponding ribbon width is plotted in Fig. 2a. A clear inverse proportionality is revealed, in excellent quantitative agreement with our calculations based on the Hubbard model (equation (1)) considering hydrogen-saturated and relaxed edges, as well as first-principles DFT calculations[16]. The theoretical data points correspond to three classes of armchair ribbons: with $3n$, $3n + 1$ and $3n + 2$ (for $n = 1, 2, 3, \ldots$) rows of carbon dimers across their width. The measurement error and the emerging atomic-scale edge irregularities do not allow us to distinguish clearly between ribbons belonging to the $3n$ and $3n + 1$ classes; however, armchair ribbon segments with no detectable gap have been observed in a few cases, which probably belong to the $3n + 2$ class. Consequently, our measurements systematically verify the predicted mechanism of bandgap opening caused by the lateral confinement of the charge carriers in ribbons of precisely armchair edge orientation.

A strikingly different behaviour is revealed in zigzag graphene nanoribbons (Fig. 2b). For zigzag graphene nanoribbons narrower than 7 nm, we observed a fairly large bandgap of about 200–300 meV. As discussed above, the broadly predicted origin of bandgap opening in zigzag ribbons is electron–electron interactions and the predicted many-body ground state implies that the edges are magnetized[16,18,19].

However, in contrast to first principles theoretical predictions[16], the measured gap suddenly vanishes for zigzag ribbons wider than 8 nm. We attribute this discrepancy to the fact that these calculations were performed at zero temperature and without doping, whereas the experimental data was acquired at room temperature and finite doping (the corresponding shift of the Fermi level ΔE_F is about 50–100 meV; see Extended Data Fig. 2c). When the effects of temperature and doping were not taken into consideration, we found the ground state to be always semiconducting (antiferromagnetic) and no steep transition to a metallic

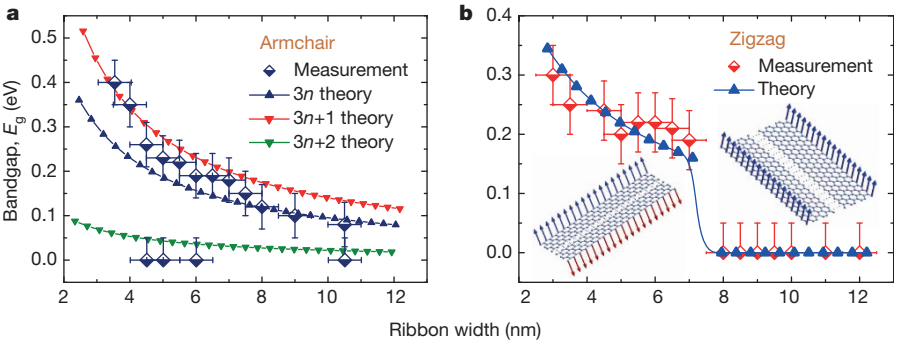

Figure 2 | Edge-specific electronic and magnetic properties of graphene nanoribbons. The bandgap measured by tunnelling spectroscopy as a function of ribbon width in armchair (**a**) and zigzag (**b**) ribbons. Armchair ribbons display a quantum confinement gap inversely proportional to their width (where $n = 1, 2, 3, \ldots$; see main text). In zigzag ribbons the band structure is governed by the emerging edge magnetism and a sharp semiconductor (antiferromagnetic) to metal (ferromagnetic) transition is revealed. Theoretical data points were calculated using the mean field Hubbard model (continuous lines are only guides to the eye). Error bars of the measured gap values originate from thermal broadening and substrate effects; error bars for ribbon widths (horizontal) arise from tip convolution effects.

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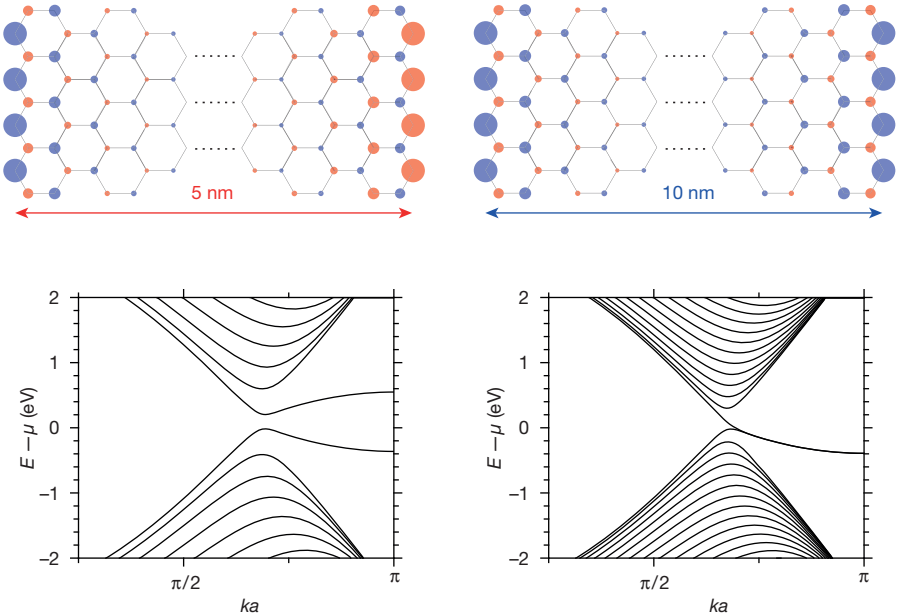

Figure 3 | Correlating electronic and magnetic properties of zigzag graphene nanoribbons. Spin density distribution (↑, blue; ↓, red) in 5-nm-wide (left) and 10-nm-wide (right) zigzag graphene nanoribbons calculated in the mean field Hubbard model for $T = 300$ K and $\Delta E_F \approx 100$ meV. The lower panels display the corresponding band structure, clearly indicating that narrow zigzag ribbons are antiferromagnetic semiconductors, whereas the wider (>8 nm) zigzag ribbons display a ferromagnetic inter-edge coupling and no bandgap. *ka* is wavenumber times the lattice constant.

state is expected. To understand the origin of the experimentally observed, sharp semiconductor–metal transition, we performed calculations based on the mean-field approximation of the Hubbard Hamiltonian that accounts for both finite temperature and doping:

$$\mathcal{H} = -\sum_{i,j} t_{ij}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + U\sum_{j}\hat{n}_{j\uparrow}\hat{n}_{j\downarrow} - \mu\hat{N} \qquad (1)$$

where the first term is the nearest-neighbour tight-binding Hamiltonian, and the second term stands for the onsite Coulomb repulsion. The effects of temperature and doping have been included using the grand canonical ensemble (third term), where μ is the chemical potential, and $\hat{N}$ is the particle number operator. The electron density and the chemical potential were determined self-consistently. (For more details on calculations see the Methods.)

The main result of our extended theoretical model is that it reproduces the observed steep semiconductor–metal transition, quantitatively accounting for both the critical ribbon width (about 7 nm) at which the transition occurs and the magnitude of the measured bandgaps, using the strength of the on-site repulsion U as the single unknown parameter (Fig. 2b). By calculating the spin density distribution for both semiconducting (<7 nm wide) and metallic (>8 nm wide) ribbons the origin of the observed semiconductor–metal transition can be identified as a transition from an antiferromagnetic (semiconducting) state, where magnetic moments on opposite ribbon edges are aligned antiparallel, to a ferromagnetic (metallic) state, with parallel spin alignment on opposite edges (Fig. 3). The metallic nature of zigzag graphene nanoribbons with ferromagnetic inter-edge coupling has already been predicted[18], together with a complex magnetic phase diagram as a function of doping. Our theoretical results are consistent with these findings in the $T \rightarrow 0$ K limit. We note that the observation of the gap opening alone in zigzag graphene nanoribbons would not unambiguously indicate edge magnetism, given that reconstructed zigzag edge configurations (for example, z_{211}) leading to semiconducting but non-magnetic zigzag graphene nanoribbons have been predicted[24]. However, only magnetic zigzag edges can account for both gap opening and the observed semiconductor–metal transition. This, together with the quantitative agreement between our calculations and experiments provide indirect but compelling evidence that magnetic order emerges on zigzag graphene edges of precisely engineered crystallographic orientation. Moreover, it is remarkable that the signature of edge magnetism can be experimentally detected at room temperature, although this is in agreement with the expected high-temperature stability of *s–p* electron magnetism[25,26], as well as experiments reporting hints of room-temperature magnetism in defective graphitic samples[5,27–29].

Since the only free parameter used in the calculations for zigzag ribbons is the magnitude of the on-site repulsion U, the best quantitative agreement with our experimental findings provides an experimental estimate for the strength of the electron–electron interaction in graphene of $U = 3.24$ eV. Theoretical works have predicted the magnitude of U to range from 2 eV to 6 eV (see, for example, ref. 30). The experimentally estimated value of U is even more reliable, because a single U value can reproduce both the magnitude of the gaps and the semiconductor–metal transition ribbon width.

That the experimental results can be quantitatively interpreted by theoretical calculations on perfect zigzag (or armchair) edges confirms the experimentally found high edge quality. However, deviations from the ideal edge structure can be present on the atomic scale. To estimate the effect of atomic-scale edge irregularities, we performed calculations on a model system having a high density of such defects (Extended Data Fig. 3). The results indicate that, even in the presence of a high defect density, for overall zigzag oriented edges the qualitative picture holds but the strength of the effect (gap size, spin polarization) decreases. To fit the experimental data, the value of electron–electron interactions estimated based on the ideal zigzag edge has to be increased from $U = 3.24$ eV to $U = 4.32$ eV.

Our findings demonstrate that graphene nanoribbons display strongly edge-orientation-specific behaviour and that engineering the crystallographic orientation of graphene edges allows us unprecedented control over both electronic and magnetic properties of graphene nanostructures. This opens the way towards the realization of electronic and spintronic devices that are based on robust quantum mechanical effects and that can operate at room temperature.

Online Content Methods, along with any additional Extended Data display items and Source Data, are available in the online version of the paper; references unique to these sections appear only in the online paper.

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Author Contributions L.T. and C.H conceived and designed the experiments. G.Z.M. performed the lithography and STM experiments. I.H. and P.V. provided the theoretical results. X.J. and C.H. performed the graphene growth experiments. Z.O. and P.N.-I. carried out preliminary experiments. L.T., P.V., I.H., G.Z.M. and L.P.B. analysed the data. L.T. wrote the paper. All the authors discussed the results and commented on the manuscript.

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METHODS

The graphene samples were grown by chemical vapour deposition on Cu foil, and transferred to a stripped gold substrate with large, atomically flat Au (111) terraces. The lithographic process and tunnelling microscopy/spectroscopy investigations were performed using a Nanoscope E Scanning Tunneling Microscope operating under ambient conditions. First, atomic-resolution STM images of the graphene lattice were acquired (bias voltage $U_{\mathrm{bias}} = 5$–50 mV and tunnelling current $I_{\mathrm{tunnel}} = 1$–2 nA), to precisely identify the zigzag and armchair directions of the graphene lattice. For patterning, a bias voltage of 2.0–2.3 V (tip negative) is applied between the tip (90% Pt, 10% Ir) and the sample while the tip is slowly (1–5 nm s^{-1}) moved along the desired cutting direction. The humidity of the cutting atmosphere (70%–75%) can be precisely controlled inside an atmospheric hood. Before the tunnelling spectroscopy measurements on ribbons, tunnelling spectra were acquired on gold to ensure that the spectra were linear and confirm that the tip was good quality and contamination-free. We acquired tunnelling spectra inside the ribbons to confirm the presence or absence of a bandgap. This method has proved more efficient and reproducible under ambient conditions than measuring strongly edge-localized peaks in the $\mathrm{d}I/\mathrm{d}V$ spectra. We define the gap of the ribbon as the width of the plateau around zero bias in tunnelling spectra (Extended Data Fig. 2b). These plateaus are not entirely flat, but have a shallow slope, which we attribute to the presence of gold substrate underneath the atomically thin graphene nanostructures. The uncertainty (error bars) in the measured gap values mainly comes from the thermal broadening and the influence of the substrate, while for ribbon widths the uncertainty is due to the STM tip convolution effects. As control experiments, both before and after measuring the I–V characteristics of a ribbon, tunnelling spectra were acquired on the graphene sheet outside the ribbons. The (nonlinear) spectra measured on the ribbons were only taken as reliable if both before and after their measurement, the spectra of the unpatterned graphene displayed the expected closely linear I–V characteristics. The tunnelling I–V characteristics of graphene and metallic graphene nanoribbons on gold substrate are close to linear owing to the contribution of the high local density of state of the Au substrate. Within (along) the ribbons the measured spectra were fairly homogeneous (Extended Data Fig. 4), apart from a few exceptions, which we attribute to local edge defects or impurities.

To interpret the experimental results we considered the Hamiltonian of equation (1). We applied the $\hat{n}_{i\sigma} = \hat{n}_{i\sigma} - <\hat{n}_{i\sigma}> + <\hat{n}_{i\sigma}>$ identity and neglected the term that contains the fluctuations. Quantum Monte Carlo simulations including fluctuations found the agreement between mean-field theory and quantum Monte Carlo to be remarkably accurate for moderate Coulomb interactions, justifying the application of mean-field theory for the description of realistic ribbon geometries[31]. We arrived at a single-particle problem, which can be diagonalized by a generalized Bogoliubov transformation in k space:

$$\mathcal{H}_{\text{mean field}} = \sum_{k,\sigma,n} (\varepsilon_{nk\sigma} - \mu)\hat{C}^{\dagger}_{n,k,\sigma}\hat{C}_{n,k,\sigma} - U\sum_{j} <\hat{n}_{j\uparrow}><\hat{n}_{j\downarrow}>$$

where $\hat{C}^{\dagger}_{n,k,\sigma}\left(\hat{C}_{n,k,\sigma}\right)$ are the transformed operators that destroy (or create) a particle with wavenumber k and spin σ in band n. The energy bands are given by $\varepsilon_{nk\sigma}$, which depends on the yet unknown electron densities and chemical potential. The densities are calculated self-consistently, while the chemical potential is determined by the conservation of the number of particles. The self-consistent iteration was stopped when the difference between the electron densities was smaller than 10^{-12}. It is known from the Hubbard-model theory that the half-filled case shows antiferromagnetic correlations for $U > 0$, and that at a certain U_c a Mott transition can take place. However, in the doped case both antiferromagnetic and ferromagnetic 'ground' states can be obtained. The solution which has the lowest free energy was accepted as the 'ground' state.

To provide accurate bandgap values for the graphene nanoribbons, we first calculated the relaxed edge geometries with hydrogen passivation of narrow (2 nm) zigzag and armchair ribbons. We used density-functional-theory-based molecular dynamics calculations within the framework of local density approximation using the VASP[32,33] simulation package. With the help of the relaxed edge coordinates of the atoms we were able to parameterize the tight-binding hopping elements in our Hubbard model. These hopping elements were applied to ribbons of various widths.

In the VASP calculations projector augmented wave pseudo-potentials[34,35] were used and the kinetic energy cut-off for the plane wave expansion was 400 eV. In all ribbon geometries the atomic positions were relaxed using the conjugate–gradient method until the forces of the atoms were reduced to 0.02 eV $\mathrm{Å}^{-1}$. We used a rectangular super-cell with $\mathrm{d}x = 40$ Å in the x direction and $\mathrm{d}y = 2.46$ Å in the y direction for the zigzag graphene nanoribbon (4.26 Å for the armchair graphene nanoribbon). The Brillouin zone was sampled using 2 k-points along the x axis and approximately $8/\mathrm{d}y$ k-points along the y axis (where $\mathrm{d}y$ is in nanometres). Vacuum layers of 20 Å in the ribbon plane and 24.6 Å in the normal direction were applied to avoid interactions between nanoribbons in different unit cells.

From the relaxed carbon–carbon distances, we computed the hopping amplitudes using the parameterization[36]

$$t(r_{ij}) = \left(\frac{r_{ij}}{a_0}\right)^{-\alpha_2} \exp\left[-\alpha_3\left(r_{ij}^{\alpha_4} - a_0^{\alpha_4}\right)\right]$$

where r_{ij} is the distance between the atoms i and j, $a_0 = 1.42$ Å (the carbon–carbon distance in the bulk), $\alpha_2 = 1.2785$, $\alpha_3 = 0.1383$ and $\alpha_4 = 3.4490$.

To estimate the maximum possible quantitative effect of edge irregularities on the magnetism, we performed the calculations presented here on a model system with edges of overall zigzag orientation, but containing the maximum amount of atomic-scale edge defects that does not completely destroy edge magnetism (Extended Data Fig. 3). We found that the qualitative picture of edge magnetism including the semiconducting (antiferromagnetic) to metallic (ferromagnetic) transition holds true for the defective zigzag edges, but the effect is substantially weakened compared to perfect edges. In particular, the calculated gap values are reduced. Therefore, the best fit to the experimental data occurs for a higher parameter value of the onsite repulsion parameter of $U = 4.32$ eV. Also, the width dependence of the gap becomes weaker, but this is still in good agreement with the measurements (Extended Data Fig. 3b) for the increased U parameter value.

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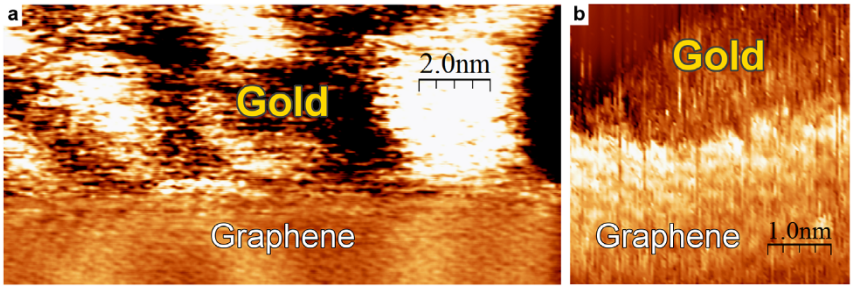

Extended Data Figure 1 | The structure of graphene edges defined by STM lithography. **a**, STM image (5 mV, 2 nA) of 15-nm-long edge segments cut by STM lithography, revealing edges that are close to atomically smooth (<5 Å edge roughness) free of detectable reconstructions, contaminations or curvature. **b**, The increased local density of states on zigzag edges observed under specific imaging conditions (200 mV, 2 nA) can be attributed to the presence of edge states that rule out the possibility of sp^3-type edge terminations (for example, di-hydrogenated edges), because no edge states are expected to occur for such edge configurations.

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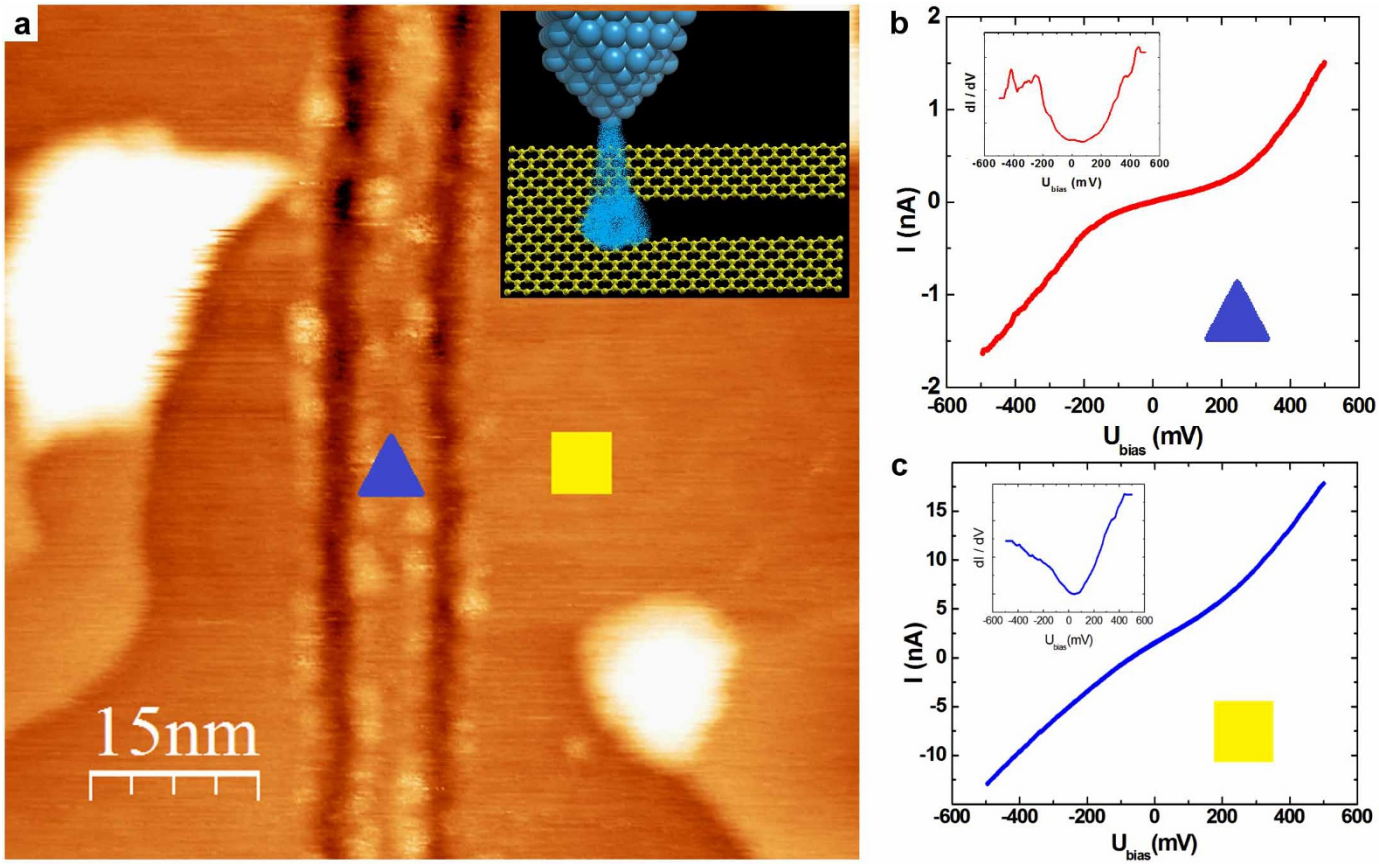

Extended Data Figure 2 | Scanning tunnelling spectroscopy of graphene nanoribbons on Au (111). Tunnelling *I*–*V* characteristics acquired on a 5-nm-wide armchair ribbon (**a**) displaying nonlinear *I*–*V* spectra corresponding to a gap of about 250 meV (**b**). Outside the ribbon a close-to-linear characteristic of the unpatterned graphene is revealed (**c**). The blue triangle and yellow square mark the positions of the corresponding tunnelling spectra. The insets show the schematics of STM lithography (**a**), and the differential tunnelling conductance (d*I*/d*V*) obtained as numerical derivatives of the measured *I*–*V* curves (**b** and **c**). The ~70-mV shift of the Dirac point (curve minimum) from the Fermi level (zero bias) observed on graphene (inset to **c**) is due to the doping from the Au(111) substrate and the ambient atmosphere.

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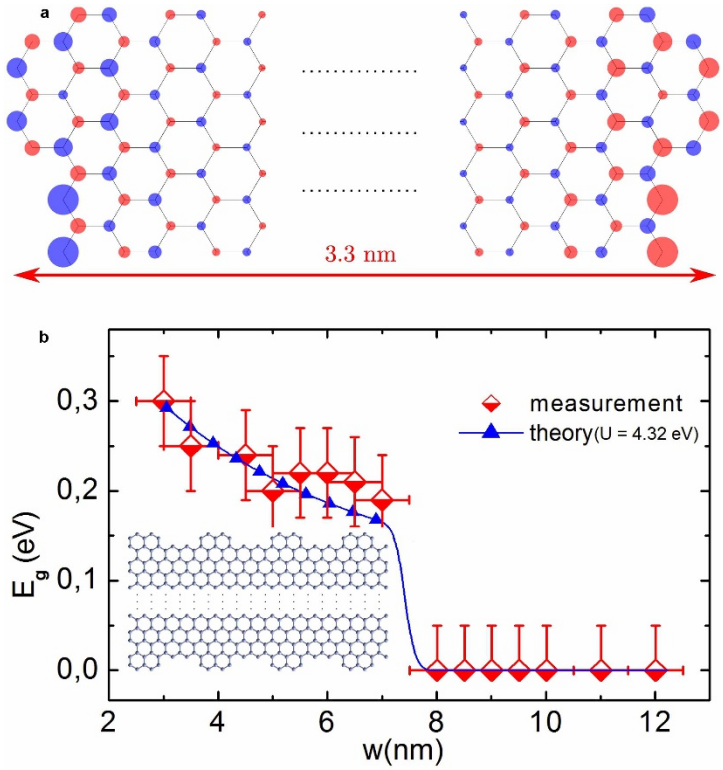

Extended Data Figure 3 | The effect of edge irregularities on edge magnetism. Calculated spin density distribution in the unit cell of a 3.3-nm-wide zigzag ribbon with a high density of atomic-scale defects, revealing the substantial decrease of the emerging spin polarization (to about a third of that of defect-free zigzag edges). The experimental width dependence can be fitted for defective ribbon edges by using higher values of the on-site repulsion parameter of $U = 4.32$ eV.

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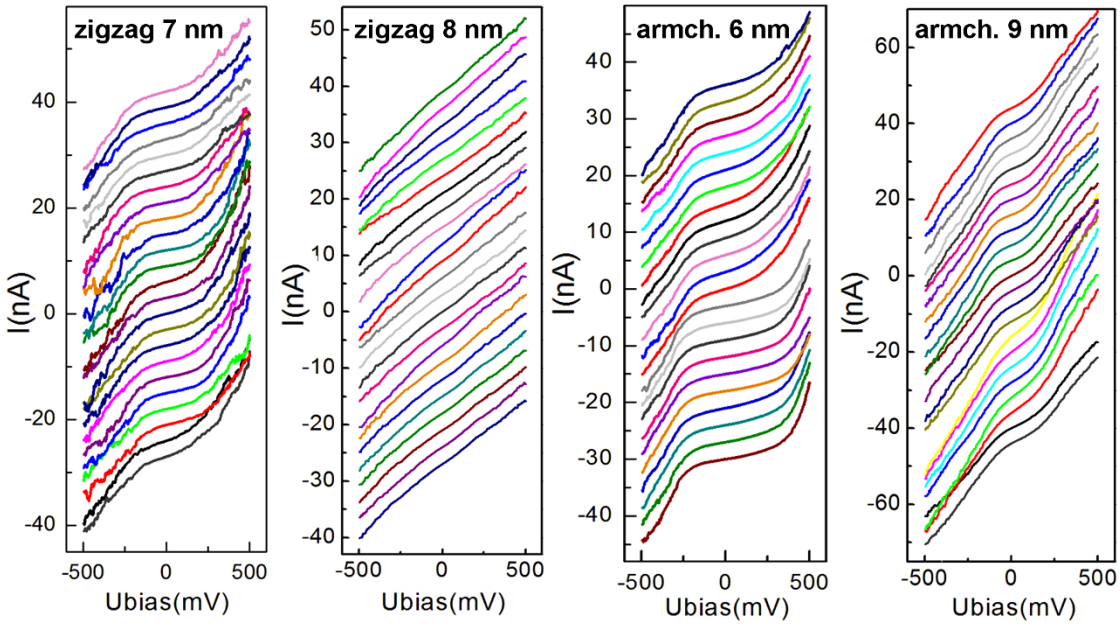

Extended Data Figure 4 | Individual tunnelling *I*–*V* spectra. Tunnelling *I*–*V* characteristics recorded on various ribbons (the spectra have been shifted along the vertical axis for clarity). Each individual spectrum was recorded as the average of ten voltage sweeps between −500 mV and +500 mV. The metallic (close to linear) or semiconducting (strongly nonlinear) nature of the ribbons is clearly apparent from the individual tunnelling *I*–*V* characteristics.

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